Being bi-invertible is a mere proposition

The Univalent Foundations Program

hott-book:equivalences.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author thm environment at source lines 394-396 and its proof environment opens on line 397, the next line; the pairing is by adjacency and the retained proof range 397-400 is the whole environment.

Editorial note. The retained counter values (chapter 4, section 3, theorem offset 1) make the counted theorem print as Theorem 4.3.2, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel thm:isprop-biinv = 4.3.2).

Editorial note. One label the proof cites lies outside the card, lem:inv-hprop (book Lemma 4.2.9, defined at line 319 of this chapter pinned file), and is declared in external_references as a non-load-bearing navigation pointer: it is counted by its own card of this lane.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: bi-invertibility is a mere proposition (author label thm:isprop-biinv, book Theorem 4.3.2).

Allowed standard prerequisites (editorial).

- Bi-invertibility. A map is bi-invertible when it has both a left inverse and a right inverse, and bi-invertibility implies the existence of a quasi-inverse.. It is the predicate shown to be a mere proposition.
- Contractibility of the two inverse types. Under a quasi-inverse both the left inverse type and the right inverse type of the map are contractible.. It makes the product of the two layers contractible.
- Products of contractible types. A product of contractible types is contractible.. It closes the argument.

Source reference. Original lemma 4.2.9: sources/hott-equivalences/pinned-original-source.tex, source lines 319–319. Book Lemma 4.2.9, contractibility of the two inverse types under a quasi-inverse, cited at source line 399. Navigation pointer only: it is counted by its own card of this lane.

Theorem 4.3.2. *For any $f : A \to B$, the type* $\mathsf{biinv}(f)$ *is a mere proposition.*

Proof. We may suppose f to be bi-invertible and show that $\mathsf{biinv}(f)$ is contractible. But since $\mathsf{biinv}(f) \to \mathsf{qinv}(f)$, by original lemma 4.2.9 in this case both $\mathsf{linv}(f)$ and $\mathsf{rinv}(f)$ are contractible, and the product of contractible types is contractible. □